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MILITARY SPECIFICATION

BOILER, STEAM, HIGH PRESSURE, WATER TUBE, FORCED CIRCULATION TYPE, AUXILIARY, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers an oil fired forced circulation water tube type boiler.

1.2 Classification. - The boilers shall be of the following classes as specified (see 6.1):

Class I - Carbon steel construction.
Class II - Nonmagnetic construction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-763 - Steel Bars, Shapes and Forgings, Corrosion Resisting.
QQ-S-766 - Steel Plates, Sheets, and Strip—Corrosion Resisting.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable, Cord and Wire Electrical (Shipboard Use).
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2174 - Controllers, Direct Current, Naval Shipboard.
MIL-C-2194 - Cable, Power, Electrical Reduced Diameter Type, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-I-2819 - Insulation Block, Thermal.
MIL-B-3242 - Boiler Compound, Navy.
MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-J-5624 - Jet Fuel, Grades JP-4 and JP-5.
MIL-Q-9858 - Quality Program Requirements.
MIL-M-9868/1- Microfilming of Engineering Documents, 35MM, for Naval Ship Systems.
MIL-P-14276 - Paint, Heat-Resisting; Silicone, Aluminum.
MIL-P-15024 - Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning, Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-I-16165 - Interference Shielding, Engine Electrical Systems.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-W-16878 - Wire, Electrical Insulated, High Temperature.
MIL-F-16884 - Fuel Oil, Diesel, Marine.
MIL-I-16910 - Interference Measurement, Electromagnetic, Methods and Limits.

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MIL-M-17059 - Motors, 60-Cycle, Alternating-Current, Fractional H. P. (Shipboard Use).
MIL-M-17060 - Motors, 60-Cycle, Alternating-Current, Integral Horsepower (Shipboard Use).
MIL-M-17413 - Motors, Direct-Current, Integral HP, Naval Shipboard.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-P-17945 - Pumps, Centrifugal, Peripheral Type, Miscellaneous Naval Shipboard Service.
MIL-F-19004 - Fans, Centrifugal, Fixed and Portable, Ventilation, Naval Shipboard.

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-167 - Mechanical Vibrations Of Shipboard Equipment.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.

DRAWINGS

NAVSHIPS

810-1385850 - Gage Piping Schedule For All Services.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue, at 33rd St., New York, N. Y. 10016.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler Construction Code.

SA-285 - Carbon-Steel Plates of Flange and Firebox Qualities (Plates 2 Inch and Under).
SB-166 - Nickel-Chromium-Iron Alloy Rod, Bar and Forgings.
SB-167 - Nickel-Chromium-Iron Alloy Seamless Pipe and Tube.
SB-168 - Nickel-Chromium-Iron Alloy Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th St., New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production one boiler shall be submitted for preproduction inspection as specified in 4.2 (see 6.3).

3.2 Electrical characteristics. - The boiler and auxiliaries shall be designed to operate on the electric power supply of the rating specified (see 6.1).

3.3 Shock resistance. - The boiler and all auxiliaries shall be adequately supported on a base. Means shall be provided for connecting the base to a ship's deck or foundation. The base, all support members for boiler and for auxiliaries and hold down bolts shall be designed to withstand the shock loadings shown

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on figure 1. Shock stresses shall be considered for each direction separately and shall not be combined. Bolts stressed in shear shall be installed in holes of the following sizes:

Nominal bolt diameter	Diameter of hole
Up to and including 3/4 inch	Nominal bolt diameter plus 1/32 inch
Larger than 3/4 inch	Nominal bolt diameter plus 1/16 inch

3.3.1 Auxiliaries and other equipments rigidly supported shall not be attached in 2-structures which can deflect relative to each other, under shock loading.

3.4 Material. -

3.4.1 Cast iron shall not be used in any part of the boiler or in any part of the boiler auxiliaries. Nodular iron may be used in such parts of the boiler auxiliaries where no other substitute for cast iron is practicable provided approval of the specific application is obtained from the command or agency concerned.

3.4.2 Class I - carbon steel construction. - Except as otherwise specified herein, material used in construction of class I boilers shall conform to the American Society of Mechanical Engineers (ASME) Boiler Construction Code. Material for shell and tube sheets shall be equal to or better than ASME material SA-285, grade C, firebox quality, with respect to maximum allowable stress taken at 500°F. and weldability (see 3.6.13).

3.4.3 Class II - nonmagnetic construction. -

3.4.3.1 General requirements. - Boilers specified herein are to be installed on ships whose magnetic signature must be extremely low. Therefore great importance is attached to use of nonmagnetic materials wherever possible without involving extensive development or affecting the design characteristics. Where experience with working parts mandates the use of magnetic ferrous materials in the absence of suitable nonmagnetic substitutes, such parts shall be as widely dispersed as practicable.

3.4.3.1.1 All other parts of all components used in the boiler assembly shall be manufactured from nonmagnetic materials. Except as otherwise specified herein, material used in the construction of equipment shall be considered nonmagnetic if it has a permeability of less than 1.08 before fabrication and not greater than 2.0 after fabrication. Material used shall be fully identified on preliminary drawings (see 3.11) and on all drawings submitted by the contractor for approval after award of the contract or order (see 3.10). Unless otherwise specified herein, the boiler construction code shall apply.

3.4.3.2 Material. - The following materials shall be used in the boiler shell assembly for class II boilers.

(a) Nickel-chromium-iron alloy conforming to ASME material SB-168 for plate, sheet and strip; SB-167 for tubing; and SB-166 for bars, rods and forgings, shall be used in fabricating all parts of the pressure vessel (shell and tube assembly) and all members connected to the pressure vessel by welding. The nickel chromium iron for class II boilers shall have carbon content held to 0.08 percent maximum and all material shall be mill annealed. Cold worked and cold formed parts shall be stress relieved for 1/2 hour at 1650°F. Hot formed parts shall be stress relieved at 1650°F. for 1 hour (not less than 1/2 hour) per inch thickness.
(b) Steel, corrosion-resisting conforming to classes 304, 321, or 347, low permeability, of QQ-S-766 may be used for such applications as gas side baffles, breechings, cover plates, structural and support members and so forth provided these members do not form an integral part of the pressure vessel exposed to steam or water and provided they are not welded to the pressure vessel.
(c) Steel, corrosion-resisting, similar to class 302, low permeability, of QQ-S-763 for all forgings other than those specified in (a).

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3.4.3.3 Electrical equipment. - Electrical equipment for class II boilers shall be as follows:

(a) All items of electrical equipment shall be fabricated of nonmagnetic material, except for the following parts:
  (1) Parts required in the magnetic circuit of the particular item of equipment.
  (2) Structural parts of motors, rotary converters, contactors, relays, thermostats, and pressure switches which would entail extensive modification and redesign to permit substitution of nonmagnetic material.
(b) Electric controllers, solenoids, and switches shall have enclosures and panels of suitable nonmagnetic metals.
(c) Stray magnetic fields caused by electrical circuits shall be kept to a minimum by making the area of electric current loops as small as possible and arranging adjacent loops so that the fields will be in opposition. Current loops shall enclose no magnetic material unless essential for proper performance of equipment.
(d) In alternating current (a.c.) motors, all connections between stator winding connections shall be made as short as possible. The power leads to the stator winding shall be run through the same opening in the motor frame.

3.5 Electromagnetic interference. - The combustion control system, complete with ignition system shall be arranged to minimize electromagnetic interference, both radiated and conducted, through the use of enclosures, shielded conduits, connectors and line filters. External leads shall be shielded to reduce electromagnetic interference by shielded metallic conduit of proper diameter with spherical-seat ball and cone fittings in accordance with MIL-I-16165. All electrical connections and wiring shall be waterproof. Measurements and limits specified in MIL-I-16910 shall be used as a guide in designing for electromagnetic interference. The arrangement of electromagnetic interference and all equipment used for same shall be subject to approval by the command or agency concerned. The control circuit complete, and circuit component enclosures, wiring conduit, relay, switches, transformers, and similar equipment comprising the complete circuit shall comply with basic requirements for electric power equipments in accordance with MIL-E-917 and as specified for the individual equipment herein. Deviation from these requirements shall be submitted to the command or agency for approval.

3.6 Design. -

3.6.1 The boilers shall operate on the forced circulation principle. The circulation rate shall exceed the evaporation rate by a margin sufficient to maintain safe tube temperatures under all steaming conditions. The unit shall be oil fired and it shall include a generating tube bank or coil assembly, steam drum, accumulator or separator burner and auxiliaries, accessories and controls necessary for a complete self-container unit. No economizers, air heaters or superheaters shall be fitted to the boiler. Unless otherwise specified (see 6.1), the steam drum, accumulator or separator shall be vertical. When the boiler is secured, the heating coils shall be capable of being completely filled with water.

3.6.2 The boiler shall be designed to permit normal operation with a momentary roll up to 45 degrees, or a permanent list of 15 degrees to either side and a permanent inclination of 5 degrees fore and aft in accordance with the requirements of this specification.

3.6.3 The boiler shall be equipped for automatic burner operation. The boiler shall automatically adjust itself to changes in load, and be capable of sustained operation without surveillance.

3.6.3.1 The automatic burner shall contain the following features:

(a) Automatic electric spark ignition.
(b) Automatic modulation over turn-down range from maximum firing rate to minimum firing rate.
(c) Automatic cycling "on-off" when steaming at loads below the turn-down range.

3.6.3.1.1 The burner turn-down range shall be the maximum attainable commensurate with safe and reliable operation (see 3.6.14).

3.6.3.1.2 The boiler shall be capable of satisfactory operation during a rapid change in steam load from 25 percent to 100 percent in a period of 20 seconds without exceeding:

(a) The moisture carryover outlined in 3.6.5.2.

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(b) A variation in pressure of plus or minus 4 pounds per square inch (p.s.i.).

The above requirements also apply to a decrease in a steam load from 100 percent to 25 percent in 20 seconds.

3.6.4 The auxiliaries, accessories and control shall include the following:

(a) Burner (see 3.6.3 and 3.6.12).
(b) Automatic ignition and combustion control system (see 3.6.14).
(c) Flame failure (safety) control (combustion safeguard) (see 3.6.14).
(d) Fuel oil system (see 3.6.15).
(e) Forced draft blower and air regulating mechanism (see 3.6.16).
(f) Feedwater system (see 3.6.17 and 3.6.18).
(g) Condensate recirculating system.
(h) Steam separator.
(i) Rotary inverter to supply alternating current for automatic ignition when d.c. power supply is specified (see 3.6.14.4).
(j) Water failure and high temperature limit cut-off control (see 3.6.14 and 3.6.18).
(k) Motors and controllers as required for driven auxiliaries (see 3.6.14 and 3.6.19).
(l) Sootblower(s), as required.
(m) Safety valves, as required (see 3.6.13).
(n) Steam stop valve (see 3.6.13).
(o) Bottom blow - off valve for accumulator.
(p) Air vent valve (see 3.6.13).
(q) 2 water gages with valves, minimum visible range 10 inches, reflex separately mounted (see 3.6.13).
(r) Steam pressure gage.
(s) Pressure limit control (see 3.6.14).
(t) Combustion chamber observation and manual ignition port (swing type).
(u) Salinometer valve (see 3.6.13 and 3.6.14).

3.6.4.1 The component parts of the boiler assembly shall be arranged with respect to each other to permit easy access for servicing or for repair and replacement of parts.

3.6.4.2 The boiler breeching shall have a flanged outlet for direct connecting to a boiler uptake.

3.6.4.3 The combustion chamber shall be designed for efficient combustion with no direct flame impingement or hot spots on any portion of the heating surface or on the furnace walls. The combustion chambers shall feature either water cooled or air cooled walls, and in design and arrangement emphasis shall be placed on long life of material employed.

3.6.4.4 The boiler shall be equipped with necessary refractory, insulation and lagging consisting of a metal jacket over the insulation. The boiler casing shall be sectionalized for ease in removal.

3.6.5 The boiler shall have the following design and operating characteristics:

Pressure design - 150 p.s.i.
Pressure working (mean) - 125 p.s.i.
Pressure working (minimum) - 115 p.s.i.
Capacity, saturated steam (lbs./hr. actual evaporation at 125 p.s.i. pressure feedwater temperature 100°F.)
Rated } As specified (see 6.1)
Maximum }
Exhaust gas temperature - 650 F. (maximum at maximum capacity)
Efficiency - 75 percent (minimum at rated capacity)
Fuel burned - Fuel oil, Diesel, in accordance with MIL-F-16884 and jet fuel JP-5 in accordance with MIL-J-5624 as specified (see 6.1).

3.6.5.1 The working pressure-mean shall be approximately the mean between the automatic burner cut-in and cut-off pressure. Working pressure-minimum shall be pressure at which the boiler restarts after automatic cut-off.

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3.6.5.2 The moisture content in steam shall not exceed 1-percent when boiler is steaming at the maximum rated capacity with the maximum chloride and total solids concentration in boiler water of 550 and 1175 parts per million (p.p.m.), respectively.

3.6.5.3 Performance of the boiler shall be based on burning the following fuel as specified (see 6.1).

(a) Diesel fuel oil of the following analysis:

Carbon	85 percent
Hydrogen	14 percent
Sulphur	1 percent
Nitrogen	Nil
Oxygen and undetermined	Nil
Moisture	Trace
B.t.u. per pound	19,375

(b) JP-5 fuel in accordance with MIL-J-5624, heating value 19,500 B.t.u. per pound.

3.6.6 The physical dimensions and weight of the boiler, complete with all auxiliaries, accessories and controls shall be the minimum commensurate with reliable and efficient operation and shall not exceed the limitations, as specified (see 6.1).

3.6.7 Provisions shall be made for connecting an air supply duct direct from the weather to the forced draft blower inlet.

3.6.8 The boiler shall be insulated so that, at overload operations, the average surface temperature shall not exceed 135°F., based upon an ambient air temperature of 100°F. and a velocity of 18 feet per minute taken 1-foot away from the casing. Local hot spots due to connections and structural ties in excess of 145°F. shall be submitted to the command or agency concerned for approval. Double casing construction with air bleed through the space may be used provided that the cooling air is not discharged into machinery spaces.

3.6.9 Insulation, thermal block shall be in accordance with MIL-I-2819.

3.6.10 Two coats of aluminum paint conforming to MIL-P-14276 shall be given to the outside surfaces of the drums, the headers and the casing.

3.6.11 Except as otherwise specified herein, all boiler auxiliaries, accessories and controls shall be in accordance with best commercial practice.

3.6.12 The burner shall be a wide range mechanical pressure type, either pressure or air atomizing. It shall be capable of smokeless operation over the turn-down range. A turn-down range to 20 percent of the maximum rated capacity shall be without change of sprayer plate.

3.6.12.1 Pilot burner, if furnished, shall be electric-ignited type. The pilot fuel shall be cut off automatically at the end of the trial-for-ignition period of the main burner. The pilot flame shall be proved prior to delivery of main burner fuel.

3.6.12.2 Air pressure atomizing burners, if used, shall be equipped with an automatic cut-off device which will positively stop boiler operation in the event of air supply failure or any derangement in air supply system resulting in improper atomization. After such emergency cut-off, the boiler shall be restarted only by manually operating the push button master switch on the control panel.

3.6.12.3 An air compressor when required for an air pressure atomizing burner shall be furnished and installed within the space limitations specified for the boiler assembly complete.

3.6.13 Except as otherwise specified herein, the pressure parts shall be designed and manufactured in accordance with the latest edition of and all addenda to section 1 of the ASME Boiler and Pressure Vessel Code and as follows:

(a) All valves, including safety valves, shall conform to the requirements of the ASME Boiler Construction Code and all addenda thereto, except that the safety valves shall be cast or forged steel and all other valves shall be bronze, or cast or forged steel.

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(b) The boiler pressure vessel shall be of fusion welded construction. Welding and inspection shall be in accordance with class A-1 pressure vessels of MIL-STD-278.
(c) Unstayed dished and flanged heads with pressure on concave side shall be used for drums and headers. The heads shall be connected to drums or headers by means of a butt-joint welded on both sides.
(d) The generating tube bank circuit or tube assembly shall be completely drainable and ventable and connections shall be provided for chemical cleaning.
(e) All pressure gages shall be flush, panel mounted, gage piping to conform to Drawing 810-1385850.

3.6.14 The automatic ignition and combustion control system shall be designed to automatically modulate the burner without "on-off" cycling or change of type sprayer plates through the entire firing range when the boiler is steaming at variable loads between maximum capacity and 20 percent of maximum capacity; and to automatically cycle the burner "on-off" when the boiler is steaming at loads under the turn-down range.

3.6.14.1 The control system shall contain the following features:

(a) Automatic restart feature (following normal shut-down only).
(b) Automatic modulation.
(c) The combustion chamber shall be purged by a minimum of five air volume changes on start-up and shall not exceed 60 seconds in duration.
(d) Electric ignition (on pilot burner if used, a dual set of electrodes when igniting main burner direct).
(e) Low fire on start-up position interlock.
(f) Steam pressure limit interlock.
(g) Interlock preventing delivery of fuel to main burner until pilot flame is proved by a combustion safeguard (if pilot is furnished) or until electrode arc is proved on main burner (see 3.6.14.4). The combustion safeguard shall also monitor the main burner flame.
(h) Air pressure (safety switch) interlock in main burner circuit.
(i) Interrupted ignition circuit.
(j) Safety shut-down on ignition failure and on flame failure.

3.6.14.2 The automatic ignition and combustion control system shall be arranged to de-energize the fuel oil solenoid valve in not more than 4 seconds after starting, in case of ignition failure. Upon failure, the fuel oil solenoid valve shall close and the blower motor shall continue to operate until the combustion chamber has been purged by a minimum of 5 air volumes changes. After completion of the purging period and safety shut-down of the automatic ignition and combustion control system, manual reset shall be required prior to further automatic operation. When electric ignition is employed on main burner the arc shall be proved prior to energizing the oil solenoid valve.

3.6.14.3 Timing periods of the automatic ignition and combustion controls shall have no significant variations due to ambient temperature variations between 32° and 120°F, or due to supply voltage variations of plus or minus 15 percent of the nominal voltage. In case of failure of supply voltage the programming control shall automatically recycle and follow the prescribed sequence when power is restored.

3.6.14.4 All controls and safety devices shall be as simple as possible and shall be in accordance with MIL-E-917 and the following:

(a) Mercury tube actuated controls are not acceptable.
(b) All components designed for high shock shall be capable of withstanding type I vibration tests in accordance with MIL-STD-167.
(c) Thermoplastic material shall not be used.
(d) Relays: All relays, time delay relays, shall be high impact shockproof units in accordance with MIL-R-2033 and the following:
Type - As required
Tripping characteristics - As required
Voltage - As required
Duty - Continuous or intermittent as required
Ambient temperature - 50°C. (122°F.)

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(e) Switches: Switches and pushbuttons shall be HI shock design and shall comply with requirements of MIL-C-2212 and the following:
Voltage rating - As required
Location - Local (mounted on operating panel)
Type - Pushbutton or rotary
Ambient temperature - 50°C. (122°F.)

NOTE: Limit switches will, of necessity, have a remote location mounted near the parent machine. Their enclosure shall be dripproof.

(f) Pressure switches: All pressure switches, except modulating pressure control switches, shall be HI-shock design and shall comply with requirements of MIL-C-2212 and the following:
Voltage rating - As required
Location - Remote
Type - Pressure actuated
Ambient temperature - as required (50°C. (122°F.) minimum
Enclosure - Dripproof.

(g) Solenoid valves: Solenoid valves shall be capable of withstanding HI-shock tests in accordance with type C requirements of MIL-S-901. The solenoid coil shall be designed for continuous duty and the following:
Voltage rating - As required
Ambient temperature - 50°C. (122°F.)
Temperature rise - Shall not exceed values specified in MIL-C-2212
Enclosure - Spraytight or better
Terminal block - Required (fitted in the solenoid enclosure)
Locations - Remote

(h) Combustion safeguard (flame failure safety control): The combustion safeguard may be infrared or ultra-violet radiation responsive. The control shall monitor the pilot (if furnished) and the main flame under all firing conditions and de-energize the fuel oil solenoid valve in from 2 to 4 seconds in event of flame failure. Following this shutdown, the same sequence shall be followed as that specified for ignition failure (see 3.6.14.2). The infrared or ultra-violet eye shall be designed to discriminate between flame and hot refractory and reject undesired signals. The control assembly and the sensing cell or eye shall be tested for HI-shock in accordance with MIL-S-901 and for vibration in accordance with MIL-STD-167.

(i) Control circuit transformer: The transformer shall conform to commercial Marine standards and the following:
Volt ampere rating - As required.
Primary voltage - As required.
Secondary voltage - 110 volts single phase (a.c.).
Enclosure - Spraytight or better.
Voltage regulation - As required.
Duty - Continuous.
Terminal posts - As required.

(j) Ignition transformer: The ignition transformer shall conform to commercial Marine standards and the following:
Volt ampere rating - As required.
Primary voltage - 110 volts single phase (a.c.).
Secondary voltage - 10,000 volts (plus or minus 250 volts).
Enclosure - Spraytight or better for remote mounted units.
Mounting location - Local or remote as required.
Ambient temperature - 50°C. (122°F.).
Electromagnetic interference - As required (see 3.5).
Duty - Intermittent.

(k) Modulating motor: The modulating motor shall conform to commercial Marine standards and the following:
Timing - 15 or 30 seconds as required.
Stroke travel - as required.
Load torque (ft.-lbs.) - 150 percent of calculated load.
Voltage - As required by circuit design.
Location - Remote.
Enclosure - Dripproof.

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(l) Rotary inverter (see 3.6.14.7): The rotary inverter used to supply AC power required for the boiler control and ignition circuit shall be in accordance with MIL-M-17556 and have the following classification requirements:
Size KW - As required.
Voltage output - 110 volts.
Phase - Single phase (AC).
Service - A.
Cycle - 60 cycle.
Prime mover - D. c. motor.
Type - Revolving armature.
Enclosure - Dripproof protected.
Speed - Constant.
Duty - Continuous.
Bearings - Ball.
Ambient temperature - 50°C. (122°F.).
Insulation - A, B or H.
R. I. S. - As required.

(m) Low-water cut-off control: The low-water cut-off control shall be one of the following types:
(1) Float type
(2) Electrode - probe type

a. Float type: The float type shall be designed for flange mounting on boiler drum or shell. The float cage or surge chamber shall be removable through companion opening in drum or shell. The switch mechanism shall be magnetically operated. The float control shall be a unit which has passed HI-shock and environmental vibration requirements in accordance with MIL-S-901 and MIL-STD-167 respectively.

b. Electrode probe type: The electrode probe type shall be designed for flange mounting on boiler drum or shell. The probes shall be located and arranged to form an isolated circuit, ungrounded. Each probe shall be mounted independently. Probes, complete with cage or surge chamber shall be removable through companion opening in drum or shell. The electrodes shall be insulated with teflon. A ground indicating circuit shall be wired into the cut-off branch of the control circuit. Arrangement shall be similar to figure 2 and switches and indicators shall be located on the control operating panel.

c. Should the boiler manufacturer desire to use a low water cut-off control other than the float or electronic probe types, specific approval must be obtained from the Naval Ship Engineering Center (NAVSEC), Washington, D. C.

(n) Motor controllers: Electric controllers (motor starters) shall be in accordance with the following:
(1) They shall be mounted remote to the boiler control circuit enclosure.
(2) MIL-C-2174 for D. C. controllers.
(3) MIL-C-2212 for A. C. controllers.
(4) The controllers shall conform to the following classification requirements:
Master switch - Remote, 3-element push button "Start", "Stop", and "Emergency".
Service - A.
Ratings - As required.
Enclosure - Dripproof.
Operation - Magnetic.
Type - Resistance for d. c. units; across line for a. c. units.
Function - Motor starting.
Duty - Continuous.
Protection - Overload protection with emergency run feature and low voltage protection.
Performance - Automatic or semi-automatic as required.
Ambient temperature - 50 C.

(o) Console or control enclosure: The enclosure shall be designed in accordance with class I type, dripproof, requirements of MIL-E-2036.

(p) Electrical wire cable and connectors shall conform to MIL-C-915, MIL-C-2194, MIL-C-5015 and MIL-W-16878.

(q) Means shall be provided for continuous determination of boiler water level or boiler water flow and shall be approved by NAVSEC, Washington, D. C.

3.6.14.5 Following shutdown for any reason, provision shall be made to insure that manual restart can only be made at low fire position. After safety shutdown of the burner, it shall be restarted by manually operating a reset button.

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3.6.14.6 The controls shall be mounted in a dripproof enclosure (see 3.6.14.4). The enclosure may be a console type or mounted on an operating panel consolidating all indicators, lights and pushbuttons thereon. The service of each pushbutton shall be marked. Particular care shall be taken in design and arrangement of all control equipment to insure safety of operating personnel from danger of electric shock. Electric motor controllers (starters) shall be arranged for separate mounting. The main control circuit transformer shall be located outside the control circuit enclosures.

3.6.14.7 Automatic control, except for motor starters, shall be suitable for operation with 110 volts a.c. Suitable transformer shall be included on the operating panel if the available a.c. power supply is other than 110 volts a.c. A rotary inverter shall be included if available power supply is direct current (see 3.2).

3.6.15 The fuel oil system shall include the following: fuel oil strainer, fuel oil pump, fuel oil filter (with nondisposable element), relief valve, solenoid oil valve, metering oil valve, as required, supply and return line pressure gages, and interconnecting valves and fittings.

3.6.15.1 Fuel oil strainers and filters shall be the duplex type so that no interruption in flow can occur when cleaning. The shipbuilder shall provide a 75 micron strainer adjacent to the JP-5 fuel tank in the supply line to the JP-5 auxiliary boiler fuel pump but prior to the duplex filter.

3.6.15.2 The fuel oil pump shall be of the self-priming type and shall have a minimum suction lift of 12 feet. The pump shall be capable of supplying all requisite oil and shall have characteristics suitable for operation of the burner furnished. The pump shall have a direct coupled transmission and a separate motor drive (motor shall have no other loading.) The pump shall be either gear, rotary or axial piston type. The bearings shall be located externally with provision made for grease or oil lubrication. Materials for rotors shall be of a nonwearable type. The pump shall be designed to operate for 10,000 hours with the output pressure not to drop 10 p.s.i. below that for which it is designed. The pump shall be tested apart from the boiler to assure compliance with desired operating characteristics for a minimum of 250 hours on diesel fuel and 250 hours on JP-5 fuel.

3.6.16 Forced draft blower. - The forced draft blower shall be designed to supply all requisite air to the boiler, when steaming at variable loads through its entire range, at a pressure of at least 3.0 inches of water in excess of the maximum pressure required to overcome all resistance of the burner and boiler between the air intake of the blower and the gas outlet of the boiler, and with a negative pressure of 3/10 inch water at the suction end. The blower wheel (fan) shall be in satisfactory static and dynamic balance. The forced draft blower shall be tested apart from the boiler to assure compliance with the desired operating characteristics for a minimum of 100 hours. The interior of the blower casing shall be smooth and free from obstructions, such as bolt heads, nuts and pipe fittings. The following construction features shall apply:

(a) The wheel shall be keyed on the shaft.
(b) Provisions shall be made for 2 set screws in the hub - 90 degrees apart.
(c) One set screw shall be recessed in the shaft for positioning axially.
(d) One set screw shall be located over the key.
(e) Frictionless type bearings shall be employed and the shaft shall be provided with a shoulder or collar to prevent axial thrust.
(f) If practicable, the blower shall have a direct coupled drive. The maximum usable pressure of the fan shall be determined by the total pressure-volume curve in accordance with the procedures specified in MIL-F-19004.

3.6.16.1 An air regulating control shall be provided for controlling air supply to the burner in correct proportion to its firing rates. The control shall be interlocked with the metering oil valve in the fuel oil system. The linkage shall be designed to provide simultaneous manual regulation of the fuel oil and air supplies to the burner when boiler is steaming at variable loads. Means shall be provided in the linkage for manual adjustment of fuel oil-air ratio as necessary for smokeless operation of the burner throughout its entire firing range.

3.6.17 The feedwater system shall include water strainer, feed pump and recirculating pump as necessary, relief valve, pressure gage, feedwater treatment tank, and necessary piping, valves and fittings, and accessories.

3.6.17.1 The pumps shall be centrifugal or positive displacement, as required. Each pump shall be suitable for the system characteristics and shall be capable of performing its duty over the boiler operating

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range without vapor locking. Each pump shall be capable of performing its duty without overloading motor drive when pumping 40°F water. Ample drip pockets and drains shall be provided. Stuffing boxes and gland seals shall be adequate in design and hold leakage to a minimum (not to exceed 1-pound per hour). Design and construction of centrifugal pumps shall conform to MIL-P-17945 with direct coupled drive and no other loading on motor drive. Use of a positive displacement pump shall be subject to specific approval of NAVSEC, Washington, D. C. Feed pumps shall be tested apart from the boiler to assure compliance with desired operating characteristics for a minimum of 100 hours.

3.6.17.2 The feed treatment tank shall be of 1 gallon or more capacity, as approved by the command or agency concerned and shall be installed in a by-pass branch in the feed water system located between the feed pump relief valve and the feed water stop and check valves. The tank shall be valved so that feed water can be passed through the main feed water line or by-passed through the branch containing the tank. The tank shall be equipped with a funnel for adding chemicals, a vent valve, and a drain valve.

3.6.18 A feedwater regulator shall be designed to maintain a constant flow of water (feed make-up or recirculated condensate) through the tubes.

3.6.18.1 A metering system shall be valved in a by-pass branch of the feed system to permit metered treatment of water with chemicals.

3.6.18.2 Naval boiler compound in accordance with MIL-B-3242 shall be used to treat and maintain the boiler water at the following conditions:

Alkalinity - 2.5 to 3.5 equivalent p.p.m (e.p.m).
Hardness - Zero
Chloride - 15 e.p.m.

The boiler shall be capable of satisfactory operation with an open feedwater system and without the need for an oxygen scavenger. The feedwater to the boiler will meet the following conditions:

Chloride - 0.15 e.p.m. max.
Hardness - 0.20 e.p.m. max.

3.6.18.3 The boiler shall be equipped with a control device which will cut off the fuel supply to the burner in the event the flow of water is interrupted or if steam generating assembly overheats.

3.6.19 Electric motors. —Electric motors for driven auxiliaries shall be in accordance with MIL-M-17556, MIL-M-17413, MIL-M-17060 and MIL-M-17059.

3.6.19.1 The motors shall conform to the following classification requirements:

Horsepower	As required.	
Number of phases	As required.	
Speed	Constant.	
Duty	Continuous.	
Enclosure	Dripproof protected.	
Service	Navy A.	
Ambient temperature	50°C.	
Insulation	A, B, or H.	
Type	Induction.	For a.c. motors.
Winding	Squirrel cage.	For a.c. motors.
Winding	Shunt (for d.c. motors).	
Bearing	Ball.	
Design	B (for a.c. motors).	
Electromagnetic interference suppression	Required (for d.c. motors).	

3.6.20 Drives for auxiliary equipment. —Wherever practicable, the drive and the driven shall be direct coupled with flexible coupling. Belt drives may be employed subject to obtaining specific approval from the command or agency concerned. Belt drives when employed shall have the following features:

(a) Matched belts shall be furnished for multiple belt drives.
(b) Belt speeds in excess of 5,500 feet per minute are not acceptable.

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(c) Drive and driven sheaves shall be in satisfactory static or dynamic balance in accordance with internally excited vibration, type II requirements of MIL-STD-167.
(d) Axial movement in drive and driven shall be controlled by thrust bearings. Thrust bearings may be of the combined radial and thrust rolling contact type.

3.6.21 Identification plates. – Each boiler shall have an identification plate conforming to type A or B of MIL-P-15024, except that identification plates of phenolic compounds are not acceptable. The identification plates shall contain the following information:

Name and description of article.
Serial number.
Federal stock number.
Name and address of manufacturer.
Contract number.
A square of about 3/4 inch with letters "INSP" at the top for affixing Government inspector stamp.
The words "U. S. ".
Design pressure, p. s. i.
Operating pressure, p. s. i.
Hydrostatic test pressure, p. s. i.
Safety valve setting, p. s. i.
Electrical characteristics.

3.6.21.1 The stock number will be furnished to the contractor by the command or agency concerned on written request.

3.7 Repair parts. - The following types of repair parts shall be furnished as specified (see 6.1):

(a) Onboard repair parts.
(b) Stock repair parts.

3.7.1 One set of repair parts shall be furnished per boiler.

3.7.2 The set of repair parts for boiler and auxiliaries and the set of repair parts for electrical equipment shall include the following:

	Percent
(a) Boiler parts:	
Manhole gaskets (if required)	800
Handhole plates (each type and size)	100
Handhole gaskets (each type and size)	800
Boiler water gage glasses	400
Gaskets - complete sets (required for boiler and burner assembly)	800
Drive belt sets (matched sets required)	200
Safety valve internals - sets (including seats, discs and springs)	200
(b) Burner parts -	
Atomizer barrel	200
Sprayer and Orifice plates or nozzle tops	600
Metering oil valve	100
Pilot burner tips	600
(c) Blower parts -	
Bearings - set	100
(d) Feedwater strainer parts -	
Strainer baskets	200
(e) Fuel oil pump parts -	
Pump (complete assembly)	100
Pump relief valve	200
Pump shaft coupling	200
Coupling inserts	600
Seal assemblies	200
(f) Air pump or compressor (if used)	
Air pump (assembly)	100

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(g)	Feedwater pump parts	
	Shaft (without impeller)	100
	Impellers (set with nuts and keys)	100
	Pump shaft coupling	100
	Coupling inserts	600
	Packing	200
	Gaskets	200
(h)	Feedwater regulator and low water cut-off parts-	
	Magnetic float type (if used):	
	Switch assembly	100
	Float, rod and magnet	100
	Electrode probe type (if used):	
	Electrode plugs, sets	300
(i)	Electric and electronic equipment parts	
	Motors, set for each type and size in accordance with MIL-M-17413 or MIL-M-17556, as applicable for d.c. motors, including 1 armature complete with commutator and shaft	100
	Motor, set for each type and size in accordance with MIL-M-17060 or MIL-M-17059, as applicable (stator coils and rewind material shall not be included)	100
	Rotary inverter (if furnished) set for each type and size in accordance with MIL-M-17556	100
	Controller, D.C., set for each type and size in accordance with MIL-C-2174	100
	Controller, a.c., set for each type and size in accordance with MIL-C-2212	100
	Relays of each type and size, including time delay units	100
	Ignition transformers	100
	Ignition electrodes	400
	Ignition cables and connectors	400
	Combustion safeguard	100
	Vacuum tubes of each type and size	400
	Combustion safeguard sensing element	400
	Modulating motor	100
	Oil solenoid valve	100
	Oil solenoid valve coils	400
	Mechanical or electric timers or programmers	100
(j)	Circulating pump parts-	
	(1) Impeller with nut and key	100
	(2) Pump shaft coupling (if used)	100
	(3) Seal assembly	200
(k)	Sootblower parts-	
	(1) Set of valve discs, seats, pinions, and springs, as necessary	100

3.7.3 Stock repair parts. - Stock repair parts shall be furnished in accordance with MIL-P-15137.

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3.7.4 Marking for identification. - Identification marking of repair parts shall be in accordance with MIL-STD-130.

3.8 Tools. -

3.8.1 The set of tools and wrenches, special, shall be as specified herein. Each tool shall be indelibly marked to indicate the service for which intended. Special tools are defined as tools not listed in the Federal Supply Catalog. (This catalog may be consulted in the office of the Government Inspector.)

3.8.2 The number of sets of tools and wrenches, special, shall be 1 set per boiler.

3.8.3 The set of tools and wrenches, special, shall include the following:

1- Handhole cover wrench, (if used).
1- Ratchet wrench.
2- Safety valve gags.
1- Safety valve seat and set of disc templates.
1- Set of gasket templates (one for each size of handhole). The templates shall be of steel 1/16 inch thick.
1- Set of feedwater pump tools and wrench as required.
1- Set of sootblower tools and wrenches as required.
1- Tool box (metal) as required.
1- Burner cone form (metal) for use in installing plastic refractory in burner opening.
1- Burner cone sweep for use with the above form.

3.9 List of repair parts and tools and wrenches. - Repair parts list shall be in accordance with MIL-P-15137.

3.10 Drawings. -

3.10.1 Commercial drawings will be acceptable provided that satisfactory technical data is furnished. Drawing identification shall be by means of manufacturer's code identification and number. NAVSEC, Washington, D.C. or agency concerned will not assign numbers. The manufacturer shall maintain tracings of all his drawings pertaining to his equipment at his plant and shall require subcontractors to do the same. A list showing the expected dates for submission of each drawing, calculations and instruction books shall be submitted not later than 30 days after the award of the contract or order. The following detailed drawings shall be submitted showing the views indicated. Where all the information cannot be included on one drawing, more than one sheet may be submitted.

(a) Assembly and foundation plan
(1) Assembly views (front, rear and side elevations), plan view and foundation of the boiler complete with all auxiliaries, accessories and controls. All components of the assembly shall be identified by piece number and listed in the material table. Overall dimensions, locations, types and sizes of all terminal connections and space required for tube renewal shall be shown. The following data shall be tabulated:
a. Pressure designed, p.s.i.
b. Pressure operating, p.s.i.
c. Pressure safety valve setting p.s.i.
d. Pressure hydrostatic test, p.s.i.
e. Heating surface, square feet
f. Furnace volume, cubic feet
g. Evaporating capacity:
1. Rated, pounds per hour
2. Maximum, pounds per hour
h. Oil rate:
1. Rated, pounds per hour
2. Maximum, pounds per hour
i. Circulation ratio
j. Tube circuit orifice data (if required) and mass flow or velocity of water, steam and mixture in tube circuits.
k. Weight dry

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l. Weight wet.
m. Electrical characteristics.
n. Exit gas temperature at overload.
o. Efficiency at rated capacity.
p. Design rate of combustion kilo B. T. U per hour (KB/HR) per square foot of heating surface.
q. Design rate of combustion KB/HR per cubic foot of furnace volume.
r. Design rate of combustion KB/HR, per square foot of radiant heat absorbing surface.
s. Air required for atomization of fuel (cubic feet per minute (c. f. m) and p. s. i.)
t. Blower capacity (c. f. m. and pressure).
u. Feed pump capacity (gallons per minute (g. p. m) and pressure)
v. Fuel oil pump capacity (g. p. m. and pressure).

(b) Master list of material and tools
(1) List of all items and subassemblies giving number of drawing item breakdown for each subassembly is shown.
(2) List all tools furnished.
(3) The material table shall be arranged as follows:
Name.
Number required.
Material.
Material specification.
Contractor's drawing number.
Contractor's part number.
Manufacturer's name.
Manufacturer's drawing or part number.
Unit weight.
Remarks.

(c) Boiler construction and weld details -
(1) Section view of the boiler shell assembly as necessary to show arrangement and details of pressure parts, internal fittings and external fittings. Show detail views of all welds. Show method of connecting tubes. Show stress calculations for the pressure vessel.

(d) Boiler sectional views -
(1) Section (side and end) views of the boiler assembly complete with burner. Identify refractory, insulation and lagging; show quantities used and give installation instructions.

(e) Burner assembly and details -
(1) Sectional view of the boiler through the burner. Show and identify all constituent parts. Show detail views of each burner part.

(f) Boiler support foundation -
(1) Plan view, side and end elevation views of the boiler base complete with boiler support members fully dimensioned.

(g) Arrangement and details of boiler casings -

(h) Boiler wiring diagrams -
(1) Show wiring diagram and schematic wiring diagram for the boiler auxiliaries, operating controls and safety controls. Identify all symbols. Identify all components in a material table. Describe, step-by-step, the start-up, operating and securing phases for the boiler control circuit. Describe function of each safety device, how it is integrated into the system and the sequence or change in sequence for each safety shutdown.

(i) Sectional assembly views, as follows complete with material tables. -
(1) Oil pump, complete with performance curves.
(2) Water pumps, complete with performance curves.
(3) Burner-blower, complete with performance curves.
(4) Electric motors, complete with design characteristics and performance curves.
(5) Motor controllers showing design characteristics and the internal wiring circuit.
(6) Belt drive assembly, showing arrangement and all design data.
(7) Clearances of all moving parts such as bearings, fan tips, etc., for maintenance purposes.

(j) Welding drawing. - This drawing shall show all welding procedures used in the fabrication of the boiler and its componenets. All drawings showing welding shall note procedure number used and refer to this drawing.

(k) Details of tubes and coils showing details of welds and welding procedures.

3.10.1.1 Where boiler parts or components are to be identical to existing standard or master drawings a reproduction of the part or component shall not be necessary. However, references shall be made to the

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standard or master drawing which applies through means of a Certification Data Sheet which shall be supplied. Working drawings submitted for approval shall be blueprints and forwarded, two each to the Naval Ship Engineering Center, Washington, D.C. or agency concerned, the Shipbuilder, Supervisor of Shipbuilding and one copy to Ships Parts Control Center, Mechanicsburg, Pennsylvania and the Naval Ship Engineering Center, Philadelphia Division, Philadelphia, Pennsylvania. Copies of final approval drawings shall be forwarded two copies each plus one set of microfilm to the Naval Ship Engineering Center, Washington, D.C. or agency concerned, one copy (Van Dyke) to the planning shipyard, one copy (blueprint) to the ship, one copy (blueprint) to the Naval Ship Engineering Center, Philadelphia Division and one copy (Van Dyke) to the Ships Parts Control Center. The boiler manufacturer shall include microfilm of subcontractors drawings in his submission. Microfilm shall be in accordance with MIL-M-9868/1.

3.10.2 All work done prior to approval of drawings shall be entirely at the contractor's risk.

3.11 Preliminary drawings. - Each bidder shall submit preliminary drawings and data, (see 6.2), with his bid to adequately describe the equipment he proposes to furnish.

3.12 Manuals. - Manuals shall be in accordance with MIL-M-15071. The manual shall contain a maintenance check-off list showing the list of items to be examined and tests to be made and the intervals when they are to be made. Content of the manual shall be arranged in accordance with MIL-M-15071 except that parts list or list of units shall be titled "List of Components". The "List of Components" shall include major parts of the boiler and burner assemblies and all individual auxiliaries, accessories, and controls furnished with the boiler. The "List of Components" shall be arranged as follows:

Column 1 - Assembly piece number.
Column 2 - Manufacturer's part number.
Column 3 - Quantity per boiler.
Column 4 - Name in full (abbreviations shall not be used).
Column 5 - Description containing application or function, material, size, rating, manufacturer's type, or model number, catalog number.
Column 6 - Federal stock number (to be added when available).
Column 7 - Manufacturer's or contractor's drawing number.
Column 8 - Weight.

3.12.1 The list of repair parts and the list of tools and wrenches furnished shall be used as the maintenance parts list. The arrangement used for these lists shall be the same as specified for the "List of Components". In addition, illustrations as required shall be included.

3.12.2 Reduced size reproductions of the following drawings, shall be included in the manual:

(a) Assembly views of the boiler, plan view and foundation.
(b) Sectional views of the boiler.
(c) Arrangement of boiler brickwork.
(d) Arrangement and sectional views of the burner unit.
(e) Assembly and sectional views of all auxiliaries.
(f) Diagrammatic equipment wiring and elementary schematic wiring diagrams.
(g) Master drawings for motors.
(h) Master drawings for controllers.

3.12.3 Manufacturer's standard bulletins applicable to auxiliaries, accessories, and controls furnished shall be included in the technical manual where necessary for complete description of construction, method of installation, operation, adjustment, and maintenance only if one item is listed on the page. If several items are listed, a new sheet should be prepared showing only the item installed.

3.12.4 Manuals for the operation and maintenance of boilers shall conform to type II of MIL-M-15071. Five copies of the manual shall be forwarded for approval to the Naval Ship Engineering Center, Washington, D.C. The number of approved manuals to be furnished shall be as specified (see 6.1). Two copies of the final approved manual shall be forwarded to the Naval Ship Engineering Center, Washington, D.C. and one copy to the Naval Ship Engineering Center, Philadelphia Division, Philadelphia, Pennsylvania.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality control system. - The supplier shall provide and maintain a system that complies with the requirements of MIL-Q-9858. Notwithstanding the requirements of this paragraph, the supplier is in no way relieved of the final responsibility to meet the requirements of this specification.

4.2 Preproduction inspection. - Preproduction inspection shall be performed by a laboratory satisfactory to the Naval Ship Engineering Center, Philadelphia Division and shall consist of the tests specified in 4.3 and the examination of 4.4.1 (see 6.3). All tests shall be run using the fuels specified in 3.6.5.3 and the boiler water maintained as specified in 3.6.18.2.

4.3 Tests. -

4.3.1 Test agenda. - Prior to testing, a test agenda shall be prepared and submitted to the NAVSEC for approval. The agenda shall include all of the tests and requirements specified in 4.3.2, 4.3.2.1 and the following:

(a) Diagrammatic sketch of the test layout showing location for each reading to be observed.
(b) Instrumentation complete for each item listed in 4.3.2.1 giving make, model and manufacturer of instrument employed.

4.3.2 Tests shall be conducted on the preproduction unit and each production unit in accordance with table I and 4.3.2.1 through 4.4.1. The tests are required to establish compliance with the requirements of this specification and to ascertain fulfillment of the capacity and efficiency requirements and satisfactoriness of control features and safety devices. During the preproduction test steady steaming at maximum capacity test outlined in table I, data shall be taken to assure that the boiler meets the carryover requirements of 3.6.5.2.

Table I - Tests

Test	Preproduction unit	Production unit
Steady steaming at maximum capacity	8-hours consecutively	1-hour
Steady steaming at rated capacity	24-hours consecutively	1-hour
Steady steaming at 50 percent rated capacity	1-hour	1-hour
Steady steaming at minimum capacity	8-hours	1-hour
Steaming at loads modulated from 120 percent capacity to minimum capacity	8-hours consecutively	-----
Steaming at low loads "on-off" cycling	300-cycles consecutively	10-cycles consecutively

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Table I - Tests (cont'd)

Safety valve by steam	1-test	1-test
Combustion safeguard	10-cycles consecutively	5-cycles consecutively
Low water cut-off	10-cycles consecutively	5-cycles consecutively
Air-pressure cut-off	10-cycles consecutively	5-cycles consecutively
Low voltage	1-cycle	1-cycle
Chemical and physical properties of oil	1-sample	-------
Rapid load increase; 25 to 100 percent in 20 seconds (See 3.6.3.1.2)	20-cycles	-------
Rapid load decrease; 100 to 25 percent in 20 seconds (see 3.6.3.1.2)	20-cycles	-------

4.3.2.1 During the operational runs specified in table I the following data shall be obtained and recorded every 15 minutes, except for the steady steaming run on the preproduction unit where data shall be obtained and recorded once each hour. Efficiency of the boiler shall be calculated. Performance curves of the boiler shall be plotted. Data are required as follows:

(a) Oil consumption.
(b) Steam quality.
(c) Blower discharge pressure. 1/
(d) Gas pressure at exit head. 1/
(e) Boiler oil pressure.
(f) Feedwater pump discharge pressure.
(g) Blower air inlet temperature. 1/
(h) Gas temperature at exit head. 1/
(i) Boiler casing temperature. 1/
(j) Motor amperes.
(k) Gas analysis CO_2, O_2 and CO.
(l) Water level in gage glass or water flow indication.
(m) Water input or steam output.
(n) Burner plenum pressure.
(o) Blower inlet pressure or barometric pressure (as applicable). 1/
(p) Gas pressure at breeching outlet.
(q) Return line oil pressure.
(r) Steam pressure.
(s) Feedwater temperature.
(t) Gas temperature at breeching outlet.
(u) Entry and exit casing head temperature.
(v) Line voltage.
(w) Dry and wet weight.
(x) R. p. m. of each drive and driven.
(y) Total electric power consumption.

1/ Not required on the production units.

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4.3.3 Leakage. - Tubes of steam generating units of a single continuous water tube shall be subjected to either 1000 p.s.i.g. hydrostatic test or 2 times the maximum discharge pressure of the circulating or feed pump, whichever is greater. Distilled water, with sodium nitrite to prevent after rust, shall be used for hydrostatic testing. Units having parallel tube circuits shall be hydrostatically tested at a pressure 2 times the circulating or feed pump maximum discharge pressure. All other pressure parts shall be subjected to a hydrostatic pressure of 1-1/2 times the boiler design pressure. The assembled boiler shall be hydrostatically tested to 125 percent of the highest pressure any boiler part is subjected to in normal operation.

4.3.4 Circulating pumps. - All circulating pumps shall be tested under system characteristics for 250 hours.

4.3.5 Control apparatus. - All automatic control and safety device apparatus shall be inspected in accordance with the applicable specifications referenced herein.

4.3.6 Safety devices. - Safety devices shall be tested by simulating failure conditions requiring action of each safety device. All automatic control features shall be proved by short-run tests in accordance with 4.3.2.1 (not less than 15 minutes).

4.4 Examination. -

4.4.1 Boiler. - Each boiler shall be examined to verify conformance to all of the requirements of this specification, not involving tests.

4.4.2 Repair parts. - All repair parts shall be examined to verify identity, that all manufacturing operations have been performed, and that they are interchangeable with the parts installed in boilers.

4.5 Sampling and inspection of preservation and packaging. - Sampling and inspection of preservation and packaging shall be performed in accordance with MIL-P-116 to verify conformance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - After test firing at manufacturer's plant the watersides and firesides of auxiliary boilers shall be thoroughly cleaned and dried. All openings shall be sealed to accomplish a waterproof closure. Static dehumification shall be provided in accordance with method II of MIL-P-116, the amount of desiccant to be provided calculated in accordance with formula II thereof. Humidity indicator plugs shall be provided. Preservation and packaging of the boiler detached auxiliaries and accessories shall be sufficient to afford adequate protection against corrosion, deterioration, physical and mechanical damage during shipment from the supply source to the using activity until early installation, and may conform to the supplier's commercial practice when such meets these requirements. All openings shall be sealed to prevent entrance of foreign matter.

5.1.1.2 Packing. - The boiler shall be shipped assembled as a complete unit, except such auxiliaries and accessories that require separate packing for preservation, packaging and security in shipment. Such parts shall be installed by the shipyard. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

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5.1.2 Onboard repair parts and tools. - Onboard repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C in accordance with the applicable specifications as follows:

(a) Mechanical parts and tools. - MIL-R-196.
(b) Electrical parts. - MIL-E-16298 and MIL-E-17555.

Items not specifically covered therein shall be unit protected and packaged in accordance with the guide lines of MIL-P-116. Repair parts and tools shall be individually packaged, except when used in sets.

5.2 Domestic shipment and storage or overseas shipment requirements. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 The boiler and accessories shall be preserved and packaged in accordance with the level A requirements of MIL-B-3180.

5.2.1.2 Repair parts and tools. - Repair parts and tools shall be preserved and packaged level A in accordance with MIL-R-196, MIL-P-17286, MIL-E-16298 and MIL-E-17555 as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of MIL-P-116. Unless otherwise specified parts shall be individually packaged except when used in sets.

5.2.1.3 Packing. -

5.2.1.3.1 Equipment. -

5.2.1.3.1.1 Level A. - Boilers shall be packed in sheathed crates conforming to MIL-C-104.

5.2.1.3.1.1.1 Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed in overseas type wood cleated plywood, nailed wood or fiber boxes conforming to PPP-B-601, PPP-B-621 or PPP-B-636 (class 2), respectively, at the option of the contractor. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds. Single items having a gross weight exceeding 200 pounds shall be packed in boxes conforming to PPP-B-601 or PPP-B-621, modified in accordance with the applicable box specification. Boxes shall be lined with a sealed caseliner conforming to MIL-L-10547.

5.2.1.3.1.2 Level B. - Boilers shall be packed in sheathed crates conforming to MIL-C-104. Boilers having a gross weight less than 10,000 pounds may be individually packed in unsheathed crates conforming to MIL-C-3774 with the following additional requirements:

(a) The burner or front end of the boiler shall be completely enclosed in a lumber or plywood housing. The housing shall be anchored to the skid base of the crate and securely strapped to the boiler.
(b) The entire contents of the crate shall be shrouded approximately two-thirds of the way down with waterproof barrier material conforming to UU-P-271. The shrouding shall be securely fastened and arranged in a manner to prevent the formation of water pockets.

5.2.1.3.1.2.1 Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed as specified in 5.2.1.3.1.1.1. Boxes shall be of the domestic type or class. Caseliners shall not be required.

5.2.1.3.2 Repair parts and tools. - Repair parts and tools shall be packed level A or B as specified in accordance with the applicable specification specified in 5.2.1.2.

5.2.1.3.3 Anchoring, blocking, bracing and cushioning shall be in accordance with JAN-P-100 and the appendix to MIL-C-104.

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5.2.1.4 Marking. - In addition to any special markings required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.1.4.1 Unpacking instructions shall be provided and shall be sealed in waterproof envelopes which shall be tacked, stapled or otherwise securely attached to the outside of each basic equipment container in the most protected location. The words "UNPACKING INSTRUCTIONS" shall be stenciled on each container indicating location of such instructions.

5.2.1.5 Manuals. - Manuals shall be packaged and packed for the level of protection specified in accordance with MIL-M-15071.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Electric power supply available (see 3.2).
(d) Whether other than vertical steam drum is required (3.6.1).
(e) Capacity of boiler (pounds per hour) (see 3.6.5):
  (1) Rated
  (2) Maximum
(f) Fuel required (see 3.6.5 and 3.6.5.3).
(g) Limiting physical dimensions and weight of boiler assembly complete with all auxiliaries, accessories, and controls (see 3.6.6):
  (1) Overall height to the top of safety valve pad
  (2) Overall width
  (3) Depth over complete assembly
  (4) Floor space occupied by the boiler assembly complete with all auxiliaries
  (5) Dry weight of the boiler assembly complete with all auxiliaries, accessories, and controls
(h) Type of repair parts required (see 3.7).
(i) Approval authority for manuals, distribution, and number required (see 3.12.4).
(j) Selection of applicable levels of preservation, packaging, packing and marking if other than as specified in 5.1 (see 5.2).

6.2 Bid data. - Bidders should furnish with their proposals the following data in triplicate:

(a) Rated capacity of the boilers, B.t.u. per hour, rated and maximum.
(b) Heating surface, square foot.
(c) Furnace volume cubic foot.
(d) Anticipated performance data at the rated and maximum capacities (efficiency, CO_2 content and temperature of exit gas, and pounds of oil per hour).
(e) Complete description of the boiler, burner, and auxiliaries.
(f) Complete description of controls and safety devices (manufacturer's bulletins shall be attached).
(g) Weight, dry and wet, of the complete unit.
(h) Itemized list of proposed onboard repair parts.
(i) Itemized list of tools and wrenches, special.
(j) Estimated weights of onboard repair parts and tools and wrenches.
(k) Three sets of blueprints of the following preliminary drawings (see 3.11):
  (1) The outline assembly views (front, sides, and rear) of the boiler complete with auxiliaries, accessories, and controls, giving overall dimensions, locations of all piping connections, and space required for tube renewal.
  (2) Sectional elevation and drawing views of the boiler and burner assembly showing arrangement of tubes and internal fittings and giving sizes of boiler tubes, thicknesses of shell plates, and identifying refractories and insulation used.
(l) Satisfactory evidence to show that the bidder has had adequate experience in manufacture of boilers of the class and size offered in the bid. A list of actual marine installations with period of time in service shall be submitted.
(m) That right is reserved to reject bids lacking in full or in part the data required in (a) through (l).

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6.3 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 4410-N049Sh)

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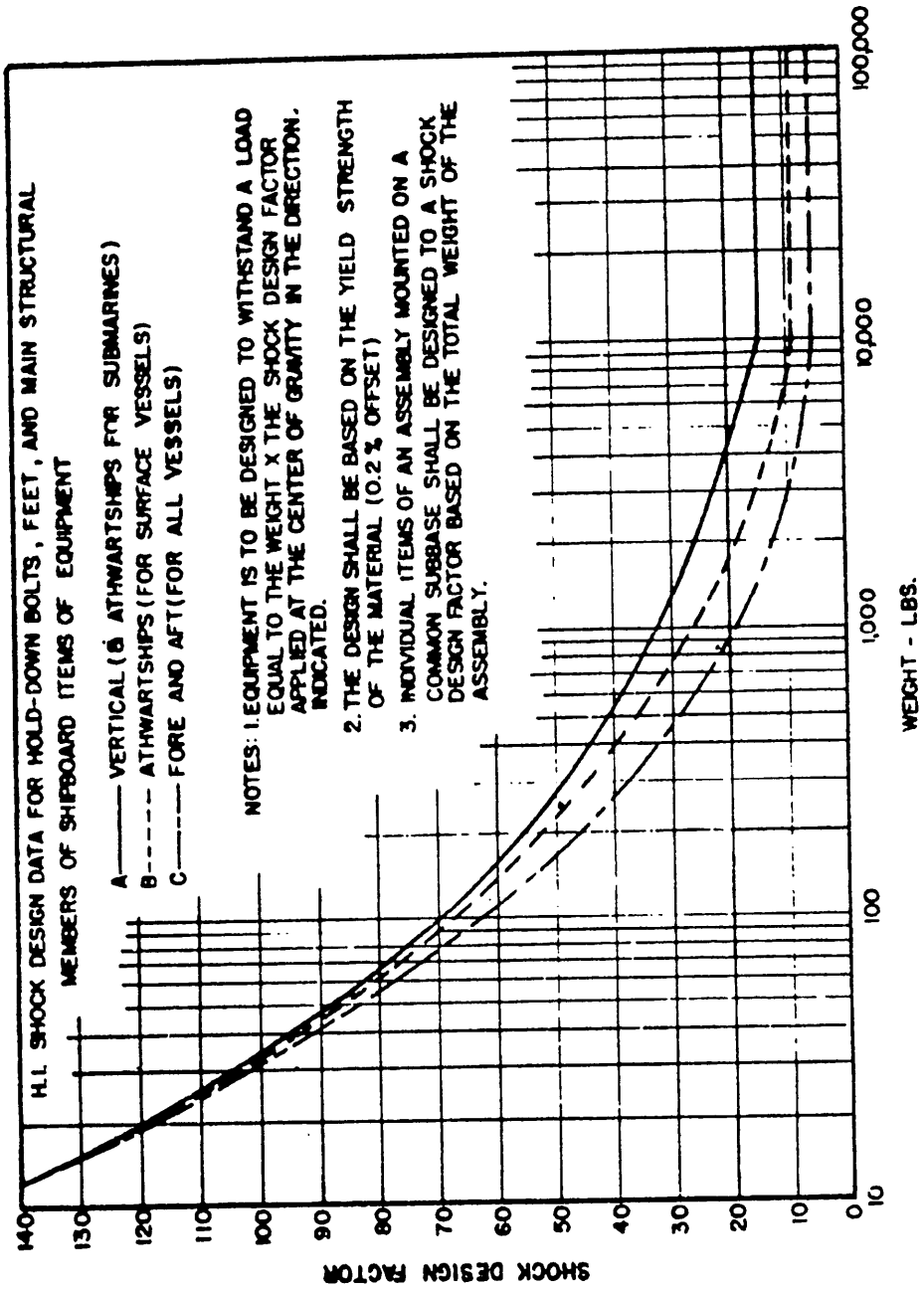


Figure 1 - Shock.

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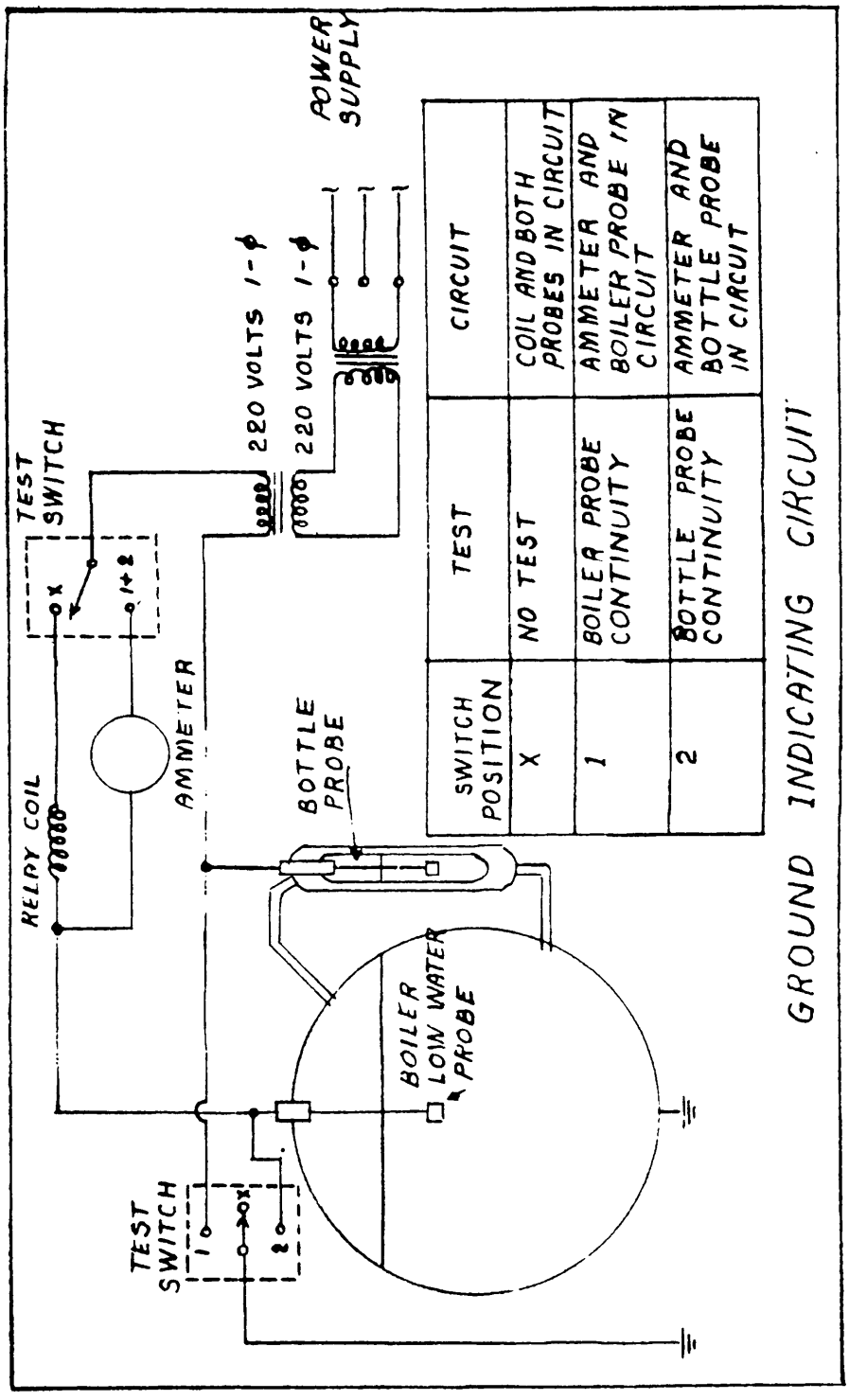


Figure 2. Ground indication circuit.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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